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Moisture hermetic guard ring for semiconductor on insulator devices

Abstract

Moisture hermetic guard ring structures for semiconductor devices, related systems, and methods of fabrication are disclosed. Such devices systems, and methods include a guard ring structure laterally surrounding semiconductor devices of a device layer and metal interconnects of an interconnect layer, the guard ring structure extending through the interconnect layer, the device layer, and a bonding layer adjacent one of the interconnect layer or the device layer the bonding layer, and contacting a support substrate coupled to the bonding layer. Such devices systems, and methods may further include via structures having the same material system as the guard ring structure and also extending through the interconnect, the device, and bonding layers and contacting a support substrate.

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Background/Summary

CLAIM OF PRIORITY (1) This application is a continuation of, and claims priority to, U.S. patent application Ser. No. 16/914,045, filed on Jun. 26, 2020 and titled “MOISTURE HERMETIC GUARD RING FOR SEMICONDUCTOR OR INSULATOR DEVICES,” which is incorporated by reference herein in its entirety.

BACKGROUND

(1) In semiconductor (or silicon) on insulator (SOI) technology, an insulator-semiconductor wafer is attached to a device wafer, for example, through silicon dioxide bonding. Such technologies offer advantages such as the ability to grind a large portion of the device wafer leaving only a thin device

wafer and thereby reducing leakage (due to the inability to control doping into the thickness of the device wafer) during eventual operation as well as providing other advantages.

(2) However, such bonding of the device wafer to the insulator-semiconductor wafer is relatively weak and exposes the bonding layer (e.g., silicon oxide to the environment). Such exposure causes the bond to become yet weaker as moisture is absorbed. During use, as a device goes through moisture exposure as well as temperature cycling, large fatigue stresses are applied to the bonding layer. In some cases, the bonding layer can be opened (e.g., beginning at an edge of the die) causing thermal and mechanical integrity loss, reduced device performance, and even failure.

(3) It is advantageous to provide more secure bonding as well as reduction or elimination of problems caused by exposure of the bonding layer. It is with respect to these and other considerations that the present improvements have been needed. Such improvements may become critical as the desire to implement SOI devices becomes more widespread.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The material described herein is illustrated by way of example and not by way of limitation in the accompanying figures. For simplicity and clarity of illustration, elements illustrated in the figures are not necessarily drawn to scale. For example, the dimensions of some elements may be exaggerated relative to other elements for clarity. Further, where considered appropriate, reference labels have been repeated among the figures to indicate corresponding or analogous elements. In the figures:

- (2) FIG. 1 is a cross-sectional view of an integrated circuit device including a guard ring structure;
- (3) FIG. 2 is another cross-sectional view of the integrated circuit device of FIG. 1;
- (4) FIG. 3 is a close up view of the integrated circuit device of FIG. 1 inclusive of a liner in the guard ring structure;
- (5) FIG. 4 is a close up view of the integrated circuit device of FIG. 1 showing a guard ring structure extending into a portion of a support substrate;
- (6) FIG. 5 is a cross-sectional view of another integrated circuit device including a guard ring structure;
- (7) FIG. 6 is a close up view of the integrated circuit device of FIG. 5;
- (8) FIG. 7 is a cross-sectional view of an integrated circuit device including a guard ring structure 107 and corresponding via structures;
- (9) FIG. 8 is another cross-sectional view of the integrated circuit device of FIG. 7;
- (10) FIG. 9 is a cross-sectional view of another integrated circuit device including a guard ring structure and corresponding via structures;
- (11) FIG. 10 is a flow diagram illustrating methods for forming integrated circuit device structures with hermetic guard ring structures;
- (12) FIGS. 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, and 21 are cross-sectional views of exemplary device structures as selected fabrication operations in the methods of FIG. 1 are performed;
- (13) FIG. 22 is an illustrative diagram of a mobile computing platform employing a device having a hermetic guard ring structure; and
- (14) FIG. 23 is a functional block diagram of a computing device, all arranged in accordance with at least some implementations of the present disclosure.

DETAILED DESCRIPTION

(15) One or more embodiments or implementations are now described with reference to the enclosed figures. While specific configurations and arrangements are discussed, it should be understood that this is done for illustrative purposes only. Persons skilled in the relevant art will recognize that other configurations and arrangements may be employed without departing from the

spirit and scope of the description. It will be apparent to those skilled in the relevant art that techniques and/or arrangements described herein may also be employed in a variety of other systems and applications other than what is described herein.

(16) Reference is made in the following detailed description to the accompanying drawings, which form a part hereof, wherein like numerals may designate like parts throughout to indicate corresponding or analogous elements. It will be appreciated that for simplicity and/or clarity of illustration, elements illustrated in the figures have not necessarily been drawn to scale. For example, the dimensions of some of the elements may be exaggerated relative to other elements for clarity. Further, it is to be understood that other embodiments may be utilized and structural and/or logical changes may be made without departing from the scope of claimed subject matter. It should also be noted that directions and references, for example, up, down, top, bottom, over, under, and so on, may be used to facilitate the discussion of the drawings and embodiments and are not intended to restrict the application of claimed subject matter. Therefore, the following detailed description is not to be taken in a limiting sense and the scope of claimed subject matter defined by the appended claims and their equivalents.

(17) In the following description, numerous details are set forth. However, it will be apparent to one skilled in the art, that the present invention may be practiced without these specific details. In some instances, well-known methods and devices are shown in block diagram form, rather than in detail, to avoid obscuring the present invention. Reference throughout this specification to “an embodiment” or “one embodiment” means that a particular feature, structure, function, or characteristic described in connection with the embodiment is included in at least one embodiment of the invention. Thus, the appearances of the phrase “in an embodiment” or “in one embodiment” in various places throughout this specification are not necessarily referring to the same embodiment of the invention. Furthermore, the particular features, structures, functions, or characteristics may be combined in any suitable manner in one or more embodiments. For example, a first embodiment may be combined with a second embodiment anywhere the particular features, structures, functions, or characteristics associated with the two embodiments are not mutually exclusive.

(18) As used in the description of the invention and the appended claims, the singular forms “a”, “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will also be understood that the term “and/or” as used herein refers to and encompasses any and all possible combinations of one or more of the associated listed items.

(19) The terms “coupled” and “connected,” along with their derivatives, may be used herein to describe structural relationships between components. It should be understood that these terms are not intended as synonyms for each other. Rather, in particular embodiments, “connected” may be used to indicate that two or more elements are in direct physical or electrical contact with each other. “Coupled” may be used to indicate that two or more elements are in either direct or indirect (with other intervening elements between them) physical or electrical contact with each other, and/or that the two or more elements co-operate or interact with each other (e.g., as in a cause and effect relationship).

(20) The terms “over,” “under,” “between,” “on”, and/or the like, as used herein refer to a relative position of one material layer or component with respect to other layers or components. For example, one layer disposed over or under another layer may be directly in contact with the other layer or may have one or more intervening layers. Moreover, one layer disposed between two layers may be directly in contact with the two layers or may have one or more intervening layers. In contrast, a first layer “on” a second layer is in direct contact with that second layer. Similarly, unless explicitly stated otherwise, one feature disposed between two features may be in direct contact with the adjacent features or may have one or more intervening features. The term immediately adjacent indicates such features are in direct contact. Furthermore, the terms “substantially,” “close,” “approximately,” “near,” and “about,” generally refer to being within +/−10% of a target value. The term layer as used herein may include a single material or multiple

materials. As used in throughout this description, and in the claims, a list of items joined by the term “at least one of” or “one or more of” can mean any combination of the listed terms. For example, the phrase “at least one of A, B or C” can mean A; B; C; A and B; A and C; B and C; or A, B and C.

(21) Semiconductor devices, device structures, apparatuses, computing platforms, and methods are described below related to semiconductor on insulator technology and, in particular, to moisture hermetic guard ring structures for semiconductor on insulator devices.

(22) As described above, it may be advantageous to provide more secure bonding and to reduce or eliminate problems caused by exposure of bonding layers in semiconductor (or silicon) on insulator (SOI) devices. As used herein, the term SOI device indicates a device formed using SOI techniques inclusive of forming devices on a device substrate or wafer, forming interconnects over the device layer, attaching the interconnect layer to a carrier substrate or wafer (via a bonding layer), and removing a bulk portion of the device substrate or wafer while leaving the device layer to expose a back side of the device layer. Traditionally, another substrate or wafer (i.e., a support substrate) is attached to the back side of the device layer (via an insulator bonding layer) and the carrier substrate or wafer is removed. Alternatively, the carrier substrate may remain as a part of the final product (i.e., as a support substrate).

(23) In either case, a trench is formed in the resultant work piece such that the trench surrounds the semiconductor devices of the device layer and the metal interconnects of the interconnect layer and such that the trench extends through the device layer, the interconnect layer, and the bonding layer and lands on or within the support substrate. Notably, the trench may be formed by sequentially etching the device layer, the interconnect layer, and the bonding layer when the support substrate is on the interconnect layer side or by etching the interconnect layer, the device layer, and the bonding layer when the support substrate is on the device layer side. The resultant trench is then filled with a guard ring structure that laterally surrounds the devices of the device layer, the metal interconnects of the interconnect layer and an internal portion of the bonding layer. Therefore, the guard ring structure laterally surrounds a die including an integrated circuit. In some embodiments, the discussed techniques are performed at the wafer level, which is diced to provide each die with a guard ring as discussed.

(24) The guard ring structure may include a liner (e.g., a polymer to aid in adhesion and/or smooth serrated sidewalls of the trench), a barrier layer (e.g., a metal oxide to block diffusion of a fill material and to again aid in adhesion of the fill material), and a fill material. In some embodiments, the fill material is copper. Notably, the guard ring structure blocks moisture from the bonding layer within the perimeter of the guard ring structure and thereby provides a hermetic guard ring to protect the bonding layer (and other materials) from becoming weaker due to moisture absorption and related problems due to the moisture caused during temperature cycling of the device in operation. Furthermore, the guard ring structure, largely due to the fill material, provides improved bonding energy and strength in the bond between the support substrate and the device layer/interconnect layer work piece relative to use of the bonding layer alone.

(25) In addition, during such trench and guard ring structure formation, via structures, having the same materials as the guard ring may be formed interspersed within the region defined by the perimeter of the guard ring structure. Such via structures (e.g., through semiconductor or silicon vias, TSVs) may be employed for heat dissipation, grounding, or other uses. In particular, in embodiments where the support substrate is adjacent the interconnect layer (and therefore opposite the device layer), the thinned device layer may produce heat that is not dissipated by a bulk semiconductor material (which was removed) and may not have access to a ground plane. In such contexts, the guard ring structure may provide heat dissipation and/or connection to the ground plane. In addition or in the alternative, the via structures may provide such heat dissipation and/or ground plane connection. Such via structures may also be employed when the support substrate is adjacent the device layer (and therefore opposite the interconnect layer) as discussed further herein.

(26) The disclosed techniques, device structures, devices, and systems herein provide a moisture blocking guard ring structure through a bonding layer and surrounding a die. As discussed, the guard ring structure may be formed in wafer-to-wafer fabrication work flows and provide a die that is resilient to delamination at the bond interface while providing the device advantages that SOI techniques afford. The resultant hermetic guard ring structure provides bond layer integrity to withstand thermo-mechanical stress coming from the package and other conditions of use for improved reliability.

(27) FIG. 1 is a cross-sectional view of an integrated circuit device **100** including a guard ring structure **107** and FIG. 2 is another cross-sectional view of integrated circuit device **100** including guard ring structure **107**, arranged in accordance with at least some implementations of the present disclosure. As shown, FIG. 2 provides a view taken along the A-A plane in FIG. 1. FIG. 1, in turn, provides a view taken along the B-B plane in FIG. 2. Integrated circuit device **100** may also be characterized as a device structure or simply a device. As shown in FIG. 1, device **100** includes a device layer **101** including a number of semiconductor devices **111** (as illustrated with respect to an outlined device region) within an interior region **122** defined by guard ring structure **107** and in a region **123** outside of guard ring structure **107**. It is noted that the portions of device layer **101** (and other layers) within region **122** are discontinuous with those portions outside of region **122** and in region **123**. With reference to FIG. 2, region **122** is within a perimeter **201** defined by guard ring structure **107** and region **123** is outside of perimeter **201** defined by guard ring structure **107**.

Perimeter **201** may be any suitable shape such as rectangular, square (as shown), or other. Notably, as used herein the term ring indicates a surrounding of a region as defined by perimeter **201** but does indicate a circular or rounded shape of guard ring structure **107** or any elements thereof.

(28) Returning to FIG. 1, device layer **101** may include any number and type of semiconductor devices **111** formed within a semiconductor substrate material. In some examples, device layer **101** includes a semiconductor material such as monocrystalline silicon (Si), germanium (Ge), silicon germanium (SiGe), a III-V materials based material (e.g., gallium arsenide (GaAs)), a silicon carbide (SiC), a sapphire (Al.sub.2O.sub.3), or any combination thereof. Semiconductor devices **111** within device layer **101** may include transistors, memory devices, capacitors, resistors, optoelectronic devices, switches, or any other active or passive electronic devices. Notably, no operable semiconductor devices are within region **123**, although dummy structures may be formed in region **123** during the fabrication of semiconductor devices **111**.

(29) Device layer **101**, as discussed further herein, is formed by removing bulk substrate material from a substrate in which and/or on which device layer **101** is formed. Device layer **101** may have any suitable thickness (in the z-dimension) such as a thickness in the range of 5 to 20 nm, a thickness in the range of 10 to 25 nm, or a thickness in the range of 20 to 50 nm, or larger. In any case, device layer **101** is thin and fragile and requires support by a support substrate **106** during manufacture and in use.

(30) Device **100** includes an interconnect layer **102** on device layer **101** such that interconnect layer **102** includes metal interconnects **103** interspersed among dielectric material **104**. As with semiconductor devices **111**, metal interconnects **103** are within region **122** and absent region **123** although dummy structures and package level interconnects may be provided in region **123**. Metal interconnects **103** may include any metal materials such as copper, aluminum, etc. and metal interconnects **103** electrically couple devices **111** to one another and/or to device level interconnects for connection to an outside package, package substrate, motherboard, etc. For example, metal interconnects **103** may include alternating layers of metal lines and vias that become progressively larger moving away from device layer **101**.

(31) In some embodiments, interconnect layer **102** further includes an interconnect guard ring **113** formed during the formation of metal interconnects **103** and having the same materials as metal interconnects **103**. Interconnect guard ring **113** laterally surrounds metal interconnects **103**. As used herein, the term laterally surrounds indicates the interior items and/or layer or portion thereof are

surrounded along the plane of the layer that the items are part of but not outside of that plane. In the illustration of FIG. 1, the term lateral indicates a direction in the x-y plane. As shown, although interconnect guard ring **113** laterally surrounds metal interconnects **103**, interconnect guard ring **113** does not laterally surround device layer **101** nor any other layers of device layer **101** as devices **111**, device layer **101**, and other layers (discussed below) are not laterally aligned with interconnect guard ring **113** and indeed extend outside of an outer edge **124** of interconnect guard ring **113**. Although illustrated herein as inclusive of interconnect guard ring **113**, in some embodiments, interconnect guard ring **113** is absent.

(32) Device **100** further includes a passivation layer **112** on interconnect layer **102**. As shown, passivation layer **112** includes a portion within region **122** and a portion outside of region **122** and in region **123**. Passivation layer **112** may include any suitable material or materials to protect interconnect layer **102** such as polymer materials. Passivation layer **112** may have any suitable thickness (e.g., in the z-dimension) such as 1 to 4 microns. In some embodiments, passivation layer **112** may be characterized as included in interconnect layer **102**. Device **100** includes a bonding layer **105** on passivation layer **112** (or interconnect layer **102**). Bonding layer **105** bonds passivation layer **112** (or interconnect layer **102** in the absence of passivation layer **112**) to a support substrate **106**.

(33) Support substrate **106** may be any suitable material or material such as any of those discussed with respect to the substrate of device layer **101**. For example, support substrate **106** may include monocrystalline silicon, germanium, silicon germanium, a III-V materials based material (e.g., gallium arsenide), a silicon carbide, a sapphire, or any combination thereof. In some embodiments, the substrate of device layer **101** and support substrate **106** are silicon substrates. In some embodiments, support substrate **106** has a thickness (in the z-dimension) in the range of 500 to 750 microns, a thickness in the range of 600 to 850 microns, or a thickness in the range of 800 to 950 microns. In some embodiments, bonding layer **105** is formed by providing an oxide (e.g., SiO_x) on one or both of passivation layer **112** and a surface of support substrate **106** (with oxidation of both being advantageous), bringing them in close contact such that they bond via van der Waals forces (e.g., at room temperature), and annealing to provide covalent bonding. However, any suitable technique or techniques for permanent bonding may be used. As shown, as with device layer **101**, interconnect layer **102**, and passivation layer **112**, a portion of bonding layer **105** is within region **122** and a portion is outside of region **122** and in region **123**.

(34) As discussed, device **100** includes guard ring structure **107** laterally surrounding semiconductor devices **111** of device layer **101**, metal interconnects **103** of interconnect layer **102**, a portion of passivation layer **112**, and a portion of bonding layer **105** such that guard ring structure **107** defines region **122** within guard ring structure **107** and region **123** outside of guard ring structure **107**. As discussed, guard ring structure **107** laterally surrounds devices **111** of device layer **101**, metal interconnects **103** of interconnect layer **102**, a portion of passivation layer **112**, and a portion of bonding layer **105** such that guard ring structure **107** is lateral (in the x-y plane) and around such components.

(35) Furthermore, guard ring structure **107** extends through device layer **101**, interconnect layer **102**, passivation layer **112**, and bonding layer **105** and is in contact with support substrate **106**. As used herein, the term extends through with respect to a component indicates the component penetrates both sides of a layer and is surrounded (at least partially) by the layer. For example, interconnect guard ring **113** extends through a surface **117** of interconnect layer **102** but terminates at (and therefore does not extend through) a surface **114** of passivation layer **112** and a surface **115** of device layer **101**. In contrast, guard ring structure **107** extends through device layer **101** such that it extends through both surface **115** of device layer **101** and a surface **116** of device layer **101** opposite surface **115**, extends through interconnect layer **102** including surface **117**, extends through passivation layer **112** including surface **114** and a surface **118** of passivation layer **112** opposite surface **114**, and bonding layer **105** including both surfaces thereof, and terminates at

support substrate **106**. In the example of FIG. **1**, guard ring structure **107** terminates at surface **110** of support substrate **106**. In some embodiments, guard ring structure **107** may extend through surface **110** and terminate within the bulk material of support substrate **106** as discussed with respect to FIG. **4** herein below.

(36) Guard ring structure **107** provides a hermetic seal, improved bond strength, and/or improved mechanical properties for device **100**. Notably, with reference to FIG. **2**, an edge **203** of device **100** may be exposed to moisture and in implementation, moisture taken on by bonding layer **105** (or other layers) is blocked by guard ring structure **107**. For example, edge **203** includes edges of device layer **101**, interconnect layer **102**, passivation layer **112**, bonding layer **105**, and support substrate **106**, which may be established during a dicing operation. In some embodiments, such edges are substantially aligned horizontally and share an edge in the z-dimension). Without guard ring structure **107**, such moisture causes reduction in interfacial adhesion strength between bonding layer **105** and support substrate **106** and/or between bonding layer **105** and passivation layer **112** and can ultimately cause delamination and failure under thermo-mechanical stress in use. Furthermore, guard ring structure **107** provides improved bond strength with support substrate **106** as well as other improved mechanical properties such as capability to take on load during mechanical stress.

(37) As shown, guard ring structure **107** may include a barrier layer **109** and a fill material **108** such that fill material **108** is within barrier layer **109**. Guard ring structure **107** may also include a liner as discussed below such that a barrier layer **109** is within the liner and fill material **108** is, in turn, within barrier layer **109**. In some material systems, guard ring structure **107** includes only fill material **108**. Fill material **108** may include any suitable material that provides a hermetic seal, improved bond strength, and/or improved mechanical properties for device **100**. In some embodiments, fill material **108** is a metal. For example, fill material **108** may include copper, aluminum, tungsten, titanium, or any combination thereof. In an embodiment, fill material **108** is copper having a purity of not less than 99% by weight.

(38) Barrier layer **109** may also include any suitable material that provides, for example, improved adhesion for fill material **108**, diffusion blocking for fill material **108**, and/or improvements for guard ring structure **107** in terms of the previously discussed hermetic seal, improved bond strength, and/or improved mechanical properties. In some embodiments, barrier layer **109** is a metal oxide or combinations of metal oxides such as titanium oxide or tantalum oxide. In some embodiments, barrier layer **109** includes titanium oxide (i.e., titanium and oxygen) and fill material **108** includes copper.

(39) FIG. **3** is a close up view of integrated circuit device **100** taken at region **121** and inclusive of a liner in guard ring structure **107**, arranged in accordance with at least some implementations of the present disclosure. As shown in FIG. **3**, guard ring structure **107** may further include a liner **303**. Liner **303** may include any suitable material such as a polymer material for improved adhesion and/or to smooth serrated sidewalls of a trench in which guard ring structure **107** is formed. For example, as discussed further herein, a patterned trench may be formed by removing patterned portions of device layer **101**, interconnect layer **102**, passivation layer **112**, and bonding layer **105**. Such material removal may be performed using etch or other techniques and may provide a trench having serrated sidewalls or sidewalls that are otherwise not smooth. Application of liner **303** may advantageously provide smooth sidewalls, fill potential void spaces, and otherwise improve subsequent fill operations.

(40) Notably, guard ring structure **107** may include barrier layer **109** only, liner **303** only, or barrier layer **109** and liner **303** in addition to fill material **108**. Guard ring structure **107** has a thickness or height, h , in the z-dimension (inclusive of any materials in the implemented material system) substantially defined by the thickness or heights of device layer **101**, interconnect layer **102**, passivation layer **112**, and bonding layer **105**. For example, guard ring structure **107** may have a height in the range of 15 to 20 microns, a height in the range of 18 to 22 microns, or a height in the

range of 20 to 25 microns. The lateral width, w, of guard ring structure **107** (inclusive of any materials in the implemented material system) taken between sidewalls **305** of guard ring structure **107** may be any suitable width to provide the desired hermetic seal, improved bond strength, and/or improved mechanical properties. In some embodiments, guard ring structure **107** has a width between sidewalls **305** in the range of 3 to 5 microns, a width between sidewalls **305** in the range of 4 to 6 microns, or a width between sidewalls **305** in the range of 4.5 to 5.5 microns. Notably, widths of larger than about 5 to 6 microns may be employed but offer reducing returns in the discussed advantages.

(41) FIG. **4** is a close up view of integrated circuit device **100** taken at region **121** and showing guard ring structure extending into a portion of support substrate **106**, arranged in accordance with at least some implementations of the present disclosure. Although not illustrated in FIG. **4**, and in other contexts herein, liner **303** may be employed in any guard ring structure **107** discussed herein. With reference to FIGS. **1** and **3**, in some embodiments guard ring structure **107** lands on surface **110** of support substrate **106** such that guard ring structure **107** does not substantially penetrate into a bulk material of support substrate **106**. For example, an etch selectively may exist between support substrate **106** and bonding layer **105** such that the etch of bonding layer **105** does not substantially etch support substrate **106** and a landing on surface **110** may be provided.

(42) In some embodiments, as shown in FIG. **4**, guard ring structure **107** extends within a portion **401** of support substrate **106** such that guard ring structure **107** extends to a depth (in the z-dimension), d, of support substrate **106**. Such techniques may provide for improved desired hermetic seal, improved bond strength, and/or improved mechanical properties with respect to landing on surface **110**. In some embodiments, portion **401** may be removed via an additional substrate (e.g., semiconductor or silicon) etch after the etch of bonding layer **105**. Guard ring structure **107** may extend into support substrate **106** by any suitable dimension such as a depth in the range of 50 to 200 nm, a depth in the range of 200 to 500 nm, a depth in the range of 500 to 1,000 nm, or more. In some embodiments, Guard ring structure **107** may extend into support substrate **106** at a depth of not less than 200 nm.

(43) As discussed, bonding layer **105** and support substrate **106** are provided adjacent to interconnect layer **102** in some embodiments. In other embodiments, bonding layer **105** and support substrate **106** are provided adjacent to device layer **101**.

(44) FIG. **5** is a cross-sectional view of an integrated circuit device **500** including guard ring structure **107**, arranged in accordance with at least some implementations of the present disclosure. Notably, FIG. **2** also provides a view taken along the A-A plane in FIG. **5**. Integrated circuit device **500** may also be characterized as a device structure or simply a device. Like numerals or components in integrated circuit device **500** with respect to device **100**, and herein throughout, may have any characteristics discussed with reference thereto. As shown in FIG. **5**, device **500** includes device layer **101** inclusive of semiconductor devices **111** within region **122** defined by guard ring structure **107** and extending to region **123** outside of guard ring structure **107**. As discussed, region **122** is within perimeter **201** defined by guard ring structure **107** and region **123** is outside of perimeter **201** as defined by guard ring structure **107**.

(45) As shown, interconnect layer **102** is on device layer **101** and includes metal interconnects **103** and dielectric material **104** such that metal interconnects **103** are within region **122**. Furthermore, as discussed, interconnect layer **102** may include interconnect guard ring **113** laterally surrounding metal interconnects **103** and dielectric material **104** (but not other layers of device **500**). Device **500** further includes passivation layer **112** on interconnect layer **102** such that passivation layer **112** includes a portion within region **122** and a portion outside of region **122** and in region **123**. Furthermore, device **500** includes bonding layer **105** on device layer **101** such that bonding layer **105** bonds device layer **101** to support substrate **106**, which, as discussed, provides structural support for device layer **101** and interconnect layer **102**. In some embodiments, bonding layer **105** is formed by providing an oxide (e.g., SiO_x) on both of device layer **101** (i.e., a backside surface of

device layer **101**) and a surface of support substrate **106**, bringing them in close contact such that they bond via van der Waals forces (e.g., at room temperature), and annealing to provide covalent bonding. As shown, as with device layer **101**, interconnect layer **102**, and passivation layer **112**, a portion of bonding layer **105** is within region **122** and a portion is outside of region **122** and in region **123**.

(46) Guard ring structure **107** of device **500**, as with device **100**, laterally surrounds a portion of passivation layer **112**, metal interconnects **103** of interconnect layer **102**, semiconductor devices **111** of device layer **101**, and a portion of bonding layer **105** such that guard ring structure **107** defines region **122** within guard ring structure **107** and region **123** outside of guard ring structure **107**. Guard ring structure **107** laterally surrounds semiconductor devices **111** of device layer **101**, metal interconnects **103** of interconnect layer **102**, a portion of passivation layer **112**, and a portion of bonding layer **105** in that guard ring structure **107** is lateral and fully around (in the x-y plane) such components. Guard ring structure **107** also extends through passivation layer **112** (i.e., both surfaces **114**, **118** of passivation layer **112**), interconnect layer **102** (i.e., both opposing surfaces of interconnect layer **102**), device layer **101** (i.e., both surfaces **115**, **116** of device layer **101**), and bonding layer **105** (i.e., both opposing surfaces of bonding layer **105**) and is in contact with support substrate **106**. Notably, interconnect guard ring **113** again extends through interconnect layer **102** but terminates at surface **114** of passivation layer **112** and surface **115** of device layer **101** while guard ring structure **107** extends through both surfaces of each of passivation layer **112**, interconnect layer **102**, device layer **101**, and bonding layer **105** and terminates at or within support substrate **106**. In the example of FIG. 5, guard ring structure **107** terminates at a surface **516** of support substrate **106**. In some embodiments, guard ring structure **107** may terminate within the bulk material of support substrate **106** beyond surface **516** as discussed with respect to FIG. 6.

(47) As discussed with respect to device **100**, guard ring structure **107** in device **500** provides a hermetic seal, improved bond strength, and/or improved mechanical properties for device **100**. Furthermore, guard ring structure **107** may include fill material **108** only, barrier layer **109** and fill material **108**, liner **303** and fill material **108**, or liner **303**, barrier layer **109** and fill material **108**, as discussed herein. Any material system for guard ring structure **107** discussed herein may be employed in device **500**.

(48) FIG. 6 is a close up view of integrated circuit device **100** taken at region **521**, arranged in accordance with at least some implementations of the present disclosure. As shown in FIG. 6, guard ring structure **107** may have a height, h , and a width, w , inclusive of any suitable values such as those discussed with respect to FIG. 3. Also as shown in FIG. 6, in some embodiments, guard ring structure **107** ends or terminates at surface **516** of support substrate **106**. In other embodiments, ring structure **107** extends to a depth, d , (in the z-dimension) of support substrate **106** within a portion **601** of removed support substrate **106** at any suitable depth such as a depth in the range of 50 to 200 nm, a depth in the range of 200 to 500 nm, a depth in the range of 500 to 1,000 nm, or a depth of not less than 200 nm. Although illustrated with a material system inclusive of barrier layer **109** and fill material **108** any material system may be employed in device **500** such as fill material **108** only or liner **303**, barrier layer **109**, and fill material **108**.

(49) Notably, device **500** provides a device that may be packaged via device interconnects formed over passivation layer **112**, which are bonded to a package substrate, and a thermal solution or heat exchanger may be provided on support substrate **106**. Such structures and techniques are discussed herein with respect to FIG. 21. In contrast, with reference to FIG. 1, device **100** provides a device that may be packaged via device interconnects formed over device layer **101**, which are bonded to a package substrate, and a thermal solution or heat exchanger may be provided on support substrate **106**. Such structures and techniques are discussed herein with respect to FIG. 16. In either case, it may be advantageous to provide via structures or vias (e.g., through semiconductor or silicon vias) in concert with the discussed guard ring structure **107**.

(50) FIG. 7 is a cross-sectional view of an integrated circuit device **700** including guard ring

structure **107** and corresponding via structures **701** and FIG. **8** is another cross-sectional view of integrated circuit device **700** including guard ring structure **107** and via structures **701**, arranged in accordance with at least some implementations of the present disclosure. As shown, FIG. **8** provides a view taken along the A-A plane in FIG. **7**. FIG. **7**, in turn, provides a view taken along the B-B plane in FIG. **8**. Integrated circuit device **700** may also be characterized as a device structure or a device. Like numerals or components in integrated circuit device **500** with respect to device **100**, and herein throughout, may have any characteristics discussed with reference thereto. (51) Notably, relative to device **100**, device **700** further includes via structures **701**. Via structures **701**, or simply vias, are interspersed within region **122** such that via structures **701** may be employed in a grid pattern (as shown) or any other pattern. In some embodiments, a single via structure **701** is used. Notably, via structures **701** have the same materials (i.e., any one or more of fill material **108**, barrier layer **109**, and liner **303** as those of guard ring structure **107**. For example, guard ring structure **107** and via structures **701** are fabricated during the same processing operations as discussed further herein below. As shown, via structures **701** also extend through device layer **101**, interconnect layer **102**, passivation layer **112**, and bonding layer **105** and are in contact with support substrate **106** in the same manner as guard ring structure **107**. In the example of FIG. **7**, guard ring structure **107** and via structures **701** terminate at surface **110** of support substrate **106**. In other embodiments, guard ring structure **107** and via structures **701** terminate within the bulk material of support substrate **106** as discussed with respect to FIG. **4**.

(52) As shown, void regions **702** in metal interconnects **103** are provided in interconnect layer **102** to provide routing for via structures **701**. Via structures **701** may be used to dissipate heat from device layer **101**, to provide a ground plane connection for device layer **101**, or any other purpose. Such heat dissipation and/or ground plane connection may be provided in addition or in the alternative by guard ring structure **107**. That is, in embodiments absent via structures **701**, guard ring structure **107** may be used as a ground plane connection and/or for heat dissipation.

(53) As discussed, via structures **701** have the same materials as guard ring structure **107** inclusive of any discussed herein. In some embodiments, guard ring structure **107** and via structures **701** include a polymer liner **303**, a metal oxide barrier layer **109**, and a copper fill **108**. In some embodiments, guard ring structure **107** and via structures **701** include only a metal oxide barrier layer **109** and a copper fill **108**. Also, as shown, via structures **701** have the same height as guard ring structure **107** but, typically, a smaller width. In some embodiments, via structures **701** have a width in the range of 0.5 to 1 microns, a width in the range of 0.75 to 1.5 microns, or a width in the range of 1 to 2 microns. In some embodiments, the lower limit of the width of via structures **701** is limited by the etch used to form them.

(54) FIG. **9** is a cross-sectional view of an integrated circuit device **900** including guard ring structure **107** and corresponding via structures **701**, arranged in accordance with at least some implementations of the present disclosure. Notably, FIG. **9** also provides a view taken along the A-A plane in FIG. **8**. Integrated circuit device **900** may also be characterized as a device structure or simply a device. Like numerals or components in integrated circuit device **900** with respect to device **500**, and herein throughout, may have any characteristics discussed with reference thereto.

(55) Notably, relative to device **500**, device **900** also includes via structures **701**. Via structures **701** are interspersed within region **122** such that via structures **701** may be employed in a grid pattern (as shown) or any other pattern. In some embodiments, a single via structure **701** is used. As discussed, via structures **701** have the same materials (i.e., any one or more of fill material **108**, barrier layer **109**, and liner **303** as those of guard ring structure **107**. For example, guard ring structure **107** and via structures **701** are fabricated during the same processing operations as discussed further herein below. As shown, as with guard ring structure **107**, via structures **701** extend through passivation layer **112**, interconnect layer **102**, device layer **101**, and bonding layer **105** and are in contact with support substrate **106**. In the illustrated embodiment, guard ring structure **107** and via structures **701** terminate at surface **516** of support substrate **106**. In some

embodiments, and via structures **701** terminate within the bulk material of support substrate **106** as discussed with respect to FIG. **6**. As with device **700**, via structures **701** may be used to dissipate heat from device layer **101**, to provide a ground plane connection for device layer **101**, or any other purpose. Such heat dissipation and/or ground plane connection may be provided in addition or in the alternative by guard ring structure **107**. That is, in embodiments absent via structures **701**, guard ring structure **107** may be used as a ground plane connection and/or for heat dissipation.

(56) FIG. **10** is a flow diagram illustrating methods **1000** for forming integrated circuit device structures with hermetic guard ring structures, arranged in accordance with at least some implementations of the present disclosure. FIGS. **11**, **12**, **13**, **14**, **15**, **16**, **17**, **18**, **19**, **20**, and **21** are cross-sectional views of exemplary device structures as selected fabrication operations in methods **1000** are performed, arranged in accordance with at least some implementations of the present disclosure. It is noted that FIGS. **11**, **12**, **13**, **14**, **15**, and **16** provide a first fabrication sequence and FIGS. **11**, **12**, **13**, **17**, **18**, **19**, **20**, and **21** provide a second fabrication sequence.

(57) As shown in FIG. **10**, methods **1000** begin at operation **1001**, where a substrate wafer is received for processing and a device layer is formed within, on, and/or over the substrate wafer. The substrate wafer may be received for processing using any suitable technique or techniques and may include any materials or characteristics discussed with respect to the substrate of device layer **101**. The device layer formed within, on, and/or over the substrate wafer may include any devices discussed herein such as transistors, memory devices, capacitors, resistors, optoelectronic devices, switches, or any other active or passive electronic devices. Such devices may be performed using any suitable technique or techniques known in the art. Methods **1000** continue at operation **1002**, where an interconnect layer is formed over the device layer. The interconnect layer includes any number of layers of metal interconnects and vias interspersed among dielectric material **104**. Such metal interconnects and vias may be formed using suitable technique or techniques known in the art.

(58) FIG. **11** illustrates an example device structure **1100** including an example substrate wafer **1101**, example semiconductor devices **111**, an exemplary interconnect layer **1102**, and an exemplary passivation layer **1112**. For example, substrate wafer **1101** may be a 200 mm, 300 mm, 450 mm or any other size wafer received for processing. Substrate wafer **1101** may include any suitable substrate for forming devices **111** such as monocrystalline silicon (Si), germanium (Ge), silicon germanium (SiGe), a III-V materials based material (e.g., gallium arsenide (GaAs)), a silicon carbide (SiC), a sapphire (Al.sub.2O.sub.3), or any combination thereof. Substrate wafer **1101** may have any thickness such as a thickness in the range of 500 to 950 microns. Notably, devices **111** are formed within a small fraction (on the order of nanometers) of the thickness of substrate wafer **1101**. Furthermore, control of dopants and other characteristics of devices are difficult to control through the thickness of substrate wafer **1101** and, therefore, the following techniques may be used to eliminate negative effects corresponding to such uncontrollable regions along the depth of substrate wafer **1101**. Furthermore, substrate wafer **1101** includes a sacrificial portion **1103** (or bulk portion) that will be subsequently removed.

(59) FIG. **11** further illustrates interconnect layer **1102** and passivation layer **1112**, which may have any characteristics discussed herein with respect to interconnect layer **102** and passivation layer **112**, although interconnect layer **1102** and passivation layer **1112** are not yet patterned and separated into regions **122**, **123**. Notably, the cross-section of device structure **1100** illustrates a single example integrated circuit device or die, however device structure **1100** may include (across substrate wafer **1101** in the x-y plane) any number of such integrated circuit devices or dies.

(60) Returning to FIG. **10**, methods **1000** continue at operation **1002**, where a bonding layer is formed between a support substrate wafer and one of a device layer or an interconnect layer. That is, the support substrate wafer is attached, via a bonding layer to the interconnect layer formed at operation **1002** or to the device layer (after backside substrate removal using a sacrificial support wafer) formed at operation **1001**. FIGS. **12** and **13** are pertinent to permanently attaching the

support substrate wafer on the interconnect layer side and are discussed first. FIGS. **12**, **13**, **17**, and **18** are pertinent to permanently attaching the support substrate wafer on the device layer side are discussed subsequently. It is noted that, when applicable to attaching the support substrate wafer on the interconnect layer side, the support substrate wafer of FIGS. **12** and **13** may be characterized as permanent (as portions thereof will remain when the die packaged) while, when applicable to attaching the support substrate wafer on the interconnect layer side, the support substrate wafer of FIGS. **12** and **13** may be characterized as sacrificial or temporary (as it will be removed during subsequent fabrication).

(61) FIG. **12** illustrates an example device structure **1200**, similar to device structure **1100**, after an example support substrate wafer **1201** is coupled, via a bonding layer **1202**, to passivation layer **112** (or another layer of interconnect layer **102**). Support substrate wafer **1201** may include any material or characteristics discussed herein with respect to support substrate **106**. For example, support substrate wafer **1201** may include monocrystalline silicon, germanium, silicon germanium, a III-V materials based material (e.g., gallium arsenide), a silicon carbide, a sapphire, or any combination thereof. In some embodiments, the materials of substrate wafer **1101** and support substrate wafer **1201** are the same. In some embodiments, they are both silicon substrate wafers. In some embodiments, support substrate wafer **1201** has a thickness (in the z-dimension) in the range of 500 to 750 microns, a thickness in the range of 600 to 850 microns, or a thickness in the range of 800 to 950 microns.

(62) Support substrate wafer **1201** may be bonded, via a bonding layer **1202**, to passivation layer **1112**, which may be characterized as part of interconnect layer **1102**, or to another layer of interconnect layer **1102** using any suitable technique or techniques. In some embodiments, an oxide layer is formed on both of support substrate wafer **1201** and passivation layer **1112**. In some embodiments, support substrate wafer **1201** and substrate wafer **1101** are then brought into close proximity and bonded using van der Waals forces (e.g., the wafers are snapped together). In some embodiments, a subsequent anneal operation is performed to provide covalent bonds.

(63) FIG. **13** illustrates an example device structure **1300**, similar to device structure **1200**, after backside removal **1301** of sacrificial portion **1103** of substrate wafer **1101** to leave device layer **1302**. Backside removal **1301** of sacrificial portion **1103** may be performed using any suitable technique or techniques. In some embodiments, backside removal **1301** includes backside grind techniques, etch techniques, or combinations thereof. Notably, support substrate wafer **1201** is necessary during such processing to thin device layer **1302** to a thickness (on the order of nanometers) as required by SOI technology applications. Device layer **1302** may have any characteristics discussed herein with respect to device layer **101** although device layer **1302** is not yet patterned and separated into regions **122**, **123**.

(64) With reference to operation **1003** of FIG. **10**, such processing forms a bonding layer (e.g., bonding layer **1202**) between a support substrate wafer (e.g., support substrate wafer **1201**) and interconnect layer **1102**. In this context, bonding layer **1202** is permanent and may have any characteristics discussed with respect to bonding layer **105** although bonding layer **1202** is not yet patterned and separated into regions **122**, **123**.

(65) Discussion now turns to forming a bonding layer between a support substrate and a device layer. Such processing is also illustrated as continuing from device structure **1100** of FIG. **11** to device structure **1200** of FIG. **12**, where support substrate wafer **1201** is coupled, via bonding layer **1202**, to passivation layer **1112** (or another layer of interconnect layer **1102**). In this fabrication flow, support substrate wafer **1201** is a sacrificial support substrate and may include any material or characteristics suitable for later removal. Furthermore, bonding layer **1202** is also selected such that it is subsequently removed. For example, bonding layer **1202**, in such examples may be a UV sensitive adhesive that is releasable under UV radiation and sacrificial support substrate wafer **1201** may be translucent with respect to such UV radiation. However, any releasable bonding layer **1202** may be employed.

(66) Processing then continues as shown with respect to FIG. 13, where backside removal **1301** of the sacrificial portion of substrate wafer **1101** is performed to leave device layer **1302**. Backside removal **1301** may be performed using any suitable technique or techniques such as backside grind techniques, etch techniques, or combinations thereof, as discussed above. In this context, sacrificial support substrate wafer **1201** is again necessary to thin device layer **1302** to a thickness (on the order of nanometers) as required by SOI technology applications.

(67) Processing continues as shown with respect to FIG. 17. FIG. 17 illustrates an example device structure **1700**, similar to device structure **1300**, after support substrate wafer **1701** is coupled, via a bonding layer **1702**, to device layer **1302** (i.e., to a backside surface of device layer **101**). Support substrate wafer **1701** may include any material or characteristics discussed herein with respect to support substrate **106**. For example, support substrate wafer **1701** may include monocrystalline silicon, germanium, silicon germanium, a III-V materials based material (e.g., gallium arsenide), a silicon carbide, a sapphire, or any combination thereof. In some embodiments, the materials of substrate wafer **1101** and support substrate wafer **1701** are the same. In some embodiments, they are both silicon substrate wafers. In some embodiments, support substrate wafer **1701** has a thickness (in the z-dimension) in the range of 500 to 750 microns, a thickness in the range of 600 to 850 microns, or a thickness in the range of 800 to 950 microns.

(68) Support substrate wafer **1701** may be bonded, via a bonding layer **1702**, to device layer **1302** (i.e., to a backside surface of device layer **1302**) using any suitable technique or techniques. In some embodiments, an oxide layer is formed on one or both of support substrate wafer **1701** and the backside surface of device layer **1302**. In some embodiments, support substrate wafer **1701** and support substrate wafer **1201** are then brought into close proximity and bonded using van der Waals forces (e.g., at room temperature) and a subsequent anneal operation is performed to provide covalent bonding. Bonding layer **1702** may have any characteristics discussed herein with respect to bonding layer **105** although bonding layer **1702** is not yet patterned into regions **122**, **123**.

(69) FIG. 18 illustrates an example device structure **1800**, similar to device structure **1700**, removal of sacrificial support substrate wafer **1201** and bonding layer **1202**. Sacrificial support substrate wafer **1201** and bonding layer **1202** may be removed using any suitable technique or techniques. In some embodiments, bonding layer **1202** is releasable under UV light cure. However, any techniques may be used.

(70) Notably, with reference to operation **1003** of FIG. 10, such processing forms a bonding layer (e.g., bonding layer **1702**) between a support substrate wafer (e.g., support substrate wafer **1701**) and device layer **1302**. In this context, bonding layer **1702** is permanent (while bonding layer **1202** and support substrate wafer **1201** were temporary or sacrificial) and may have any characteristics discussed with respect to bonding layer **105** although bonding layer **1702** is not yet patterned into regions **122**, **123**.

(71) Returning to FIG. 10, methods **1000** continue at operation **1004**, where a trench is formed such that the trench laterally surrounds the semiconductor devices of the device layer and the metal interconnects of the interconnect layer and extends through the bonding layer, the interconnect layer, and the device layer to expose the support substrate wafer. The trench may be formed using any suitable technique or techniques. In some embodiments, patterning and etch techniques are employed. Furthermore, operation **1004** may be employed to pattern via openings for via structures as discussed herein.

(72) In some embodiments, a patterned layer defining the trench and/or via openings is provided over the exposed one of the device layer or the passivation layer via spin on techniques followed by lithography techniques. The defined trench and/or via openings are then etched using multi-step etch techniques to define the trench and/or via openings.

(73) First, with reference to the fabrication flow that forms the permanent bonding layer adjacent the interconnect layer, FIG. 14 illustrates an example device structure **1400**, similar to device structure **1300**, after formation of a trench **1401** and a via opening **1402**. Although illustrated with a

single via opening **1402** for the sake of clarity of presentation, any number of via openings **1402** in any pattern may be implemented. Furthermore, in some embodiments, no via openings are used. (74) In some embodiments, trench **1401** and via opening **1402** are formed by providing a patterned layer (e.g., patterned resist) on device layer **101** such that the patterned layer matches the desired patterns of trench **1401** and a via opening **1402**. Subsequently, etch operations are performed to remove, in turn, patterned portions of device layer **1302**, interconnect layer **1102**, passivation layer **1112**, and bonding layer **1202** to provide device layer **101**, interconnect layer **102**, passivation layer **112**, and bonding layer **105**. In some embodiments, a subsequent etch may be performed to provide a recess or recesses in support substrate **106** such that subsequently guard ring and/or via structures may land within a portion of support substrate **106** as discussed with respect to FIG. 4.

(75) Turning now to the fabrication flow that forms the permanent bonding layer adjacent the device layer, FIG. 19 illustrates an example device structure **1900**, similar to device structure **1800**, after formation of a trench **1901** and a via opening **1902**. As with device structure **1400**, although illustrated with a single via opening **1902** for the sake of clarity of presentation, any number of via openings **1902** in any pattern may be implemented. In some embodiments, no via openings are used. Trench **1901** and/or via opening **1902** may be formed using any suitable technique or techniques such as providing a patterned layer on passivation layer **1112** such that the patterned layer matches the desired patterns of trench **1901** and a via opening **1902**. Subsequently, etch operations are performed to remove, in turn, patterned portions of passivation layer **1112**, interconnect layer **1102**, device layer **1302**, and bonding layer **1702** to provide passivation layer **112**, interconnect layer **102**, device layer **101**, and bonding layer **105**. In some embodiments, a subsequent etch may be performed to provide a recess or recesses in support substrate **1701** such that subsequently guard ring and/or via structures may land within a portion of support substrate **1701** as discussed with respect to FIG. 6.

(76) Returning to FIG. 10, methods **1000** continue at operation **1005**, where the trench formed at operation **1004** is filled with a guard ring structure to form a guard ring structure that surrounds the semiconductor devices of the device layer and the metal interconnects of the interconnect layer, extends through the bonding layer, the interconnect layer, and the device layer, and contacts the support substrate wafer. The guard ring structure may be formed using any suitable technique or techniques. Furthermore, operation **1005** may be employed to form via structures as discussed herein.

(77) In some embodiments, formation of the guard ring structure and/or via structures includes forming, in turn, a liner layer (e.g., polymer liner), a barrier layer (e.g., a metal oxide), and a fill material, and subsequent planarization to remove excess material and provide the guard ring structure and/or via structures. In some embodiments, the liner layer is not employed and a barrier layer and fill layer are used. In some embodiments, only a fill material is used.

(78) First, with reference to the fabrication flow that forms the permanent bonding layer adjacent the interconnect layer, FIG. 15 illustrates an example device structure **1500**, similar to device structure **1400**, after formation of guard ring structure **107** and via structure **701**, which may have any characteristics discussed herein. Although illustrated with a single via structure **701** for the sake of clarity of presentation, any number of via structures **701** in any pattern may be implemented. Furthermore, in some embodiments, no via openings are used.

(79) As discussed, guard ring structure **107** and via structure **701** may be formed by first providing a conformal liner layer (not shown in FIG. 15, please refer to FIG. 3) using conformal coating or deposition techniques. The conformal liner layer conforms to trench **1401** and/or via opening **1402** in a manner similar to that of barrier layer **109**. In some embodiments, a portion of the liner layer at the bottoms of trench **1401** and via opening **1402** are removed. Barrier layer **109** is then formed using conformal deposition techniques such as atomic layer deposition (ALD). Subsequently fill material **108** is provided using bulk deposition techniques such as electroplating. Finally, portions of such materials on device layer **101** are removed using planarization techniques to provide device

structure **1500**. The liner layer, barrier layer **109**, and fill material **108** may include any material discussed herein.

(80) Turning now to the fabrication flow that forms the permanent bonding layer adjacent the device layer, FIG. **20** illustrates an example device structure **2000**, similar to device structure **1900**, after formation of guard ring structure **107** and via structure **701**. Although illustrated with a single via structure **701** for the sake of clarity of presentation, any number of via structures **701** in any pattern may be implemented. Furthermore, in some embodiments, no via openings are used. As with device structure **1500**, guard ring structure **107** and via structure **701** may be formed by first providing a conformal liner layer using conformal coating or deposition techniques, forming barrier layer **109** using conformal deposition techniques such as atomic layer deposition (ALD), and providing fill material **108** using bulk deposition techniques such as electroplating. Finally, portions of such materials on passivation layer **112** are removed using planarization techniques to provide device structure **2000**.

(81) Returning to FIG. **10**, methods **1000** continue at operation **1006**, where package level interconnects are formed over the device layer or the passivation layer, the wafer is diced to provide individual dies, each die is packaged, and thermal solutions are applied. For example, the package level interconnects are to interconnect each die to a corresponding package substrate. Such package level interconnects may be formed using any suitable technique or techniques such as metal deposition and patterning techniques, selective metallization of exposed bond pads, and so on. The wafer is then diced to provide individual dies, each having a guard ring structure as described herein.

(82) Each resultant die may then be packaged via attachment to a package substrate such as an organic printed circuit board, to an interposer, to a motherboard, or the like. In each example the package level interconnects may be formed opposite the support substrate and attachment is made opposite the support substrate. Subsequently, a heat exchanger is coupled to the support substrate. The heat exchanger may include any thermal solution such as a thermal interface material, an integrated heat spreader, a heat sink, or combinations thereof. As discussed, in some embodiments, the guard ring structure provides heat transfer from the device layer to the heat exchanger. In addition, via structures, if applicable may provide heat transfer from the device layer to the heat exchanger. Furthermore, one or both of the guard ring structure and the via structures may be attached to a ground plane to ground the device layer. Other uses of the guard ring structure and/or the via structures in the final package may be used.

(83) First, with reference to the fabrication flow that forms the permanent bonding layer adjacent the interconnect layer, FIG. **16** illustrates an example device structure **1600**, similar to device structure **1500**, after formation of package interconnects **1604**, bonding to a package substrate **1603**, application of an integrated heat spreader **1602** over device structure **1500**, and attachment of a heat sink **1601** over integrated heat spreader **1602**. As shown, of package interconnects **1604** are formed on or over device layer **101** and device layer **101** is adjacent package substrate **1603**. In some embodiments, one or both of via structure **701** and guard ring structure **107** are coupled to a ground plane **1605** via electrical trace(s) **1606**. Although illustrated with respect to a thermal solution including integrated heat spreader **1602** and heat sink **1601** as heat exchangers, any heat exchanger devices or materials may be employed to remove heat. Notably, via structure **701** and/or guard ring structure **107** provide a heat removal route from device layer **101** to support substrate **106** and, ultimately, the heat exchanger.

(84) Although illustrated with respect to bonding of a single die to a package substrate **1603**, device structure **1500** (optionally inclusive of package interconnects **1604**) may be provided in any package such as multi-chip packages.

(85) Turning now to the fabrication flow that forms the permanent bonding layer adjacent the device layer, FIG. **21** illustrates an example device structure **2100**, similar to device structure **2000**, after formation of package interconnects **1604**, bonding to package substrate **1603**, application of

integrated heat spreader **1602** over device structure **2000**, and attachment of heat sink **1601** over integrated heat spreader **1602**. As shown, in this context, package interconnects **1604** are formed on or over passivation layer **112** and interconnect layer **102**, which are adjacent package substrate **1603**. As shown, in some embodiments, one or both of via structure **701** and guard ring structure **107** are coupled to ground plane **1605** via electrical trace(s) **1606**. Although illustrated with respect to a thermal solution including integrated heat spreader **1602** and heat sink **1601** as heat exchangers, any heat exchanger devices or materials may be employed to remove heat in the context of device structure **2100**.

(86) Although illustrated with respect to bonding of a single die to a package substrate **1603**, device structure **1500** (optionally inclusive of package interconnects **1604**) may be provided in any package such as multi-chip packages.

(87) FIG. 22 is an illustrative diagram of a mobile computing platform **2200** employing a device having a hermetic guard ring structure, arranged in accordance with at least some implementations of the present disclosure. Any die or device having any hermetic guard ring structure and/or via structures and/or related characteristics discussed herein may be implemented by any component of mobile computing platform **2200**. Mobile computing platform **2200** may be any portable device configured for each of electronic data display, electronic data processing, wireless electronic data transmission, or the like. For example, mobile computing platform **2200** may be any of a tablet, a smart phone, a netbook, a laptop computer, etc. and may include a display screen **2205**, which in the exemplary embodiment is a touchscreen (e.g., capacitive, inductive, resistive, etc. touchscreen), a chip-level (SoC) or package-level integrated system **2210**, and a battery **2215**. Battery **2215** may include any suitable device for providing electrical power such as a device consisting of one or more electrochemical cells and electrodes to couple to an outside device.

(88) Integrated system **2210** is further illustrated in the expanded view **2220**. In the exemplary embodiment, packaged device **2250** (labeled “Memory/Processor” in FIG. 22) includes at least one memory chip (e.g., RAM), and/or at least one processor chip (e.g., a microprocessor, a multi-core microprocessor, or graphics processor, or the like). In an embodiment, the package device **2250** is a microprocessor including an SRAM cache memory. As shown, device **2250** may employ a die or device having any hermetic guard ring structure and/or via structures and/or related characteristics discussed herein. Packaged device **2250** may be further coupled to (e.g., communicatively coupled to) a board, a substrate, or an interposer **2260** along with, one or more of a power management integrated circuit (PMIC) **2230**, RF (wireless) integrated circuit (RFIC) **2225** including a wideband RF (wireless) transmitter and/or receiver (TX/RX) (e.g., including a digital baseband and an analog front end module further comprises a power amplifier on a transmit path and a low noise amplifier on a receive path), and a controller thereof **2235**. In general, packaged device **2250** may also be coupled to (e.g., communicatively coupled to) display screen **2205**. As shown, one or both of PMIC **2230** and RFIC **2225** may employ a die or device having any hermetic guard ring structure and/or via structures and/or related characteristics discussed herein.

(89) Functionally, PMIC **2230** may perform battery power regulation, DC-to-DC conversion, etc., and so has an input coupled to battery **2215** and with an output providing a current supply to other functional modules. In an embodiment, PMIC **2230** may perform high voltage operations. As further illustrated, in the exemplary embodiment, RFIC **2225** has an output coupled to an antenna (not shown) to implement any of a number of wireless standards or protocols, including but not limited to Wi-Fi (IEEE 802.11 family), WiMAX (IEEE 802.16 family), IEEE 802.20, long term evolution (LTE), Ev-DO, HSPA+, HSDPA+, HSUPA+, EDGE, GSM, GPRS, CDMA, TDMA, DECT, Bluetooth, derivatives thereof, as well as any other wireless protocols that are designated as 3G, 4G, 5G, and beyond. In alternative implementations, each of these board-level modules may be integrated onto separate ICs coupled to the package substrate of packaged device **2250** or within a single IC (SoC) coupled to the package substrate of the packaged device **2250**.

(90) FIG. 23 is a functional block diagram of a computing device **2300**, arranged in accordance

with at least some implementations of the present disclosure. Computing device **2300** may be found inside platform **2200**, for example, and further includes a motherboard **2302** hosting a number of components, such as but not limited to a processor **2301** (e.g., an applications processor) and one or more communications chips **2304**, **2305**. Processor **2301** may be physically and/or electrically coupled to motherboard **2302**. In some examples, processor **2301** includes an integrated circuit die packaged within the processor **2301**. In general, the term “processor” may refer to any device or portion of a device that processes electronic data from registers and/or memory to transform that electronic data into other electronic data that may be stored in registers and/or memory. Any one or more device or component of computing device **2300** may include a die or device having any hermetic guard ring structure and/or via structures and/or related characteristics discussed herein as discussed herein.

(91) In various examples, one or more communication chips **2304**, **2305** may also be physically and/or electrically coupled to the motherboard **2302**. In further implementations, communication chips **2304** may be part of processor **2301**. Depending on its applications, computing device **2300** may include other components that may or may not be physically and electrically coupled to motherboard **2302**. These other components may include, but are not limited to, volatile memory (e.g., DRAM) **2307**, **2308**, non-volatile memory (e.g., ROM) **2310**, a graphics processor **2312**, flash memory, global positioning system (GPS) device **2313**, compass **2314**, a chipset **2306**, an antenna **2316**, a power amplifier **2309**, a touchscreen controller **2311**, a touchscreen display **2317**, a speaker **2315**, a camera **2303**, and a battery **2318**, as illustrated, and other components such as a digital signal processor, a crypto processor, an audio codec, a video codec, an accelerometer, a gyroscope, and a mass storage device (such as hard disk drive, solid state drive (SSD), compact disk (CD), digital versatile disk (DVD), and so forth), or the like.

(92) Communication chips **2304**, **2305** may enables wireless communications for the transfer of data to and from the computing device **2300**. The term “wireless” and its derivatives may be used to describe circuits, devices, systems, methods, techniques, communications channels, etc., that may communicate data through the use of modulated electromagnetic radiation through a non-solid medium. The term does not imply that the associated devices do not contain any wires, although in some embodiments they might not. Communication chips **2304**, **2305** may implement any of a number of wireless standards or protocols, including but not limited to those described elsewhere herein. As discussed, computing device **2300** may include a plurality of communication chips **2304**, **2305**. For example, a first communication chip may be dedicated to shorter range wireless communications such as Wi-Fi and Bluetooth and a second communication chip may be dedicated to longer range wireless communications such as GPS, EDGE, GPRS, CDMA, WiMAX, LTE, Ev-DO, and others.

(93) As used in any implementation described herein, the term “module” refers to any combination of software, firmware and/or hardware configured to provide the functionality described herein. The software may be embodied as a software package, code and/or instruction set or instructions, and “hardware”, as used in any implementation described herein, may include, for example, singly or in any combination, hardwired circuitry, programmable circuitry, state machine circuitry, and/or firmware that stores instructions executed by programmable circuitry. The modules may, collectively or individually, be embodied as circuitry that forms part of a larger system, for example, an integrated circuit (IC), system on-chip (SoC), and so forth.

(94) While certain features set forth herein have been described with reference to various implementations, this description is not intended to be construed in a limiting sense. Hence, various modifications of the implementations described herein, as well as other implementations, which are apparent to persons skilled in the art to which the present disclosure pertains are deemed to lie within the spirit and scope of the present disclosure.

(95) In one or more first embodiments, a device comprising a device layer adjacent to an interconnect layer, the device layer comprising a plurality of semiconductor devices and the

interconnect layer comprising a plurality of metal interconnects, a bonding layer on a support substrate and adjacent one of the interconnect layer or the device layer, and a guard ring structure, wherein the guard ring structure laterally surrounds the semiconductor devices and the metal interconnects, extends through the bonding layer, the interconnect layer, and the device layer, and is in contact with the support substrate.

(96) In one or more second embodiments, further to the first embodiments, the device further comprises a second guard ring structure within the interconnect layer, wherein the second guard ring structure surrounds the metal interconnects and extends from a first surface of the interconnect layer and terminates at a first surface of the device layer, wherein the guard ring structure extends through the first surface of the interconnect layer, the first surface of the device layer, and a second surface of the device layer opposite the first surface of the device layer.

(97) In one or more third embodiments, further to the first or second embodiments, the bonding layer is on a first surface of the support substrate and the guard ring structure extends not less than 200 nm beyond the first surface into support substrate.

(98) In one or more fourth embodiments, further to any of the first through third embodiments, the bonding layer is adjacent the interconnect layer and the device further comprises a plurality of via structures within a perimeter of the guard ring structure, wherein each of the via structures extends through the bonding layer, the interconnect layer, and the device layer, and is in contact with the support substrate.

(99) In one or more fifth embodiments, further to any of the first through fourth embodiments, at least one of the plurality of via structures is to couple to a ground plane.

(100) In one or more sixth embodiments, further to any of the first through fifth embodiments, the guard ring structure comprises a barrier layer and a fill material within the barrier layer, the barrier layer comprising a metal and oxygen and the fill material comprising copper.

(101) In one or more seventh embodiments, further to any of the first through sixth embodiments, the guard ring structure further comprises a polymer liner, wherein the barrier layer is within the polymer liner.

(102) In one or more eighth embodiments, further to any of the first through seventh embodiments, the support substrate and the device layer comprise silicon and the bonding layer comprises silicon and oxygen.

(103) In one or more ninth embodiments, further to any of the first through eighth embodiments, the guard ring structure comprises a lateral width between sidewalls thereof of not less than 4 microns.

(104) In one or more tenth embodiments, further to any of the first through ninth embodiments, the guard ring structure defines a substantially rectangular or square perimeter around the semiconductor devices and the metal interconnects.

(105) In one or more eleventh embodiments, further to any of the first through tenth embodiments, the guard ring structure is laterally surrounded by outer edges of the device layer, the interconnect layer, the bonding layer, and the support substrate.

(106) In one or more twelfth embodiments, a system comprises an interconnect layer on a device layer, the device layer comprising a plurality of semiconductor devices and the interconnect layer comprising a plurality of metal interconnects, a bonding layer immediately between a support substrate and one of the interconnect layer or the device layer, a guard ring structure, wherein the guard ring structure laterally surrounds the semiconductor devices and the metal interconnects, extends through the bonding layer, the interconnect layer, and the device layer, and is in contact with the support substrate, and a heat exchanger thermally coupled to the support substrate.

(107) In one or more thirteenth embodiments, further to the twelfth embodiments, the system further comprises a second guard ring structure within the interconnect layer, wherein the second guard ring structure surrounds the metal interconnects and extends from a first surface of the interconnect layer and terminates at a first surface of the device layer, wherein the guard ring

structure extends through the first surface of the interconnect layer, the first surface of the device layer, and a second surface of the device layer opposite the first surface of the device layer.

(108) In one or more fourteenth embodiments, further to the twelfth or thirteenth embodiments, the system further comprises a plurality of package interconnects opposite the support substrate with respect to the device layer and the interconnect layer and a package substrate coupled to the package interconnects.

(109) In one or more fifteenth embodiments, further to any of the twelfth through fourteenth embodiments, the bonding layer is adjacent the interconnect layer and the system further comprises a plurality of via structures within a perimeter of the guard ring structure, wherein each of the via structures extends through the bonding layer, the interconnect layer, and the device layer, and is in contact with the support substrate.

(110) In one or more sixteenth embodiments, further to any of the twelfth through fifteenth embodiments, at least one of the plurality of via structures is coupled to a ground plane.

(111) In one or more seventeenth embodiments, further to any of the twelfth through sixteenth embodiments, the bonding layer is on a first surface of the support substrate and the guard ring structure extends not less than 200 nm beyond the first surface into support substrate.

(112) In one or more eighteenth embodiments, further to any of the twelfth through seventeenth embodiments, the guard ring structure comprises a barrier layer and a fill material within the barrier layer, the barrier layer comprising a metal and oxygen and the fill material comprising copper.

(113) In one or more nineteenth embodiments, further to any of the twelfth through eighteenth embodiments, the guard ring structure further comprises a polymer liner, wherein the barrier layer is within the polymer liner.

(114) In one or more twentieth embodiments, further to any of the twelfth through nineteenth embodiments, the support substrate and the device layer comprise silicon and the bonding layer comprises silicon and oxygen.

(115) In one or more twenty-first embodiments, further to any of the twelfth through twentieth embodiments, the guard ring structure comprises a lateral width between sidewalls thereof of not less than 4 microns.

(116) In one or more twenty-second embodiments, further to any of the twelfth through twenty-first embodiments, the guard ring structure defines a substantially rectangular or square perimeter around the semiconductor devices and the metal interconnects.

(117) In one or more twenty-third embodiments, further to any of the twelfth through twenty-second embodiments, the guard ring structure is laterally surrounded by outer edges of the device layer, the interconnect layer, the bonding layer, and the support substrate.

(118) In one or more twenty-fourth embodiments, a method of fabricating a device comprises disposing an interconnect layer over a device layer, the device layer comprising a plurality of semiconductor devices and the interconnect layer comprising a plurality of metal interconnects, forming a bonding layer on a support substrate and adjacent one of the device layer or the interconnect layer, forming a trench, wherein the trench laterally surrounds the semiconductor devices and the metal interconnects, extends through the bonding layer, the interconnect layer, and the device layer, and exposes the support substrate, and filling the trench with a guard ring structure.

(119) In one or more twenty-fifth embodiments, further to the twenty-fourth embodiments, the bonding layer is adjacent the interconnect layer and the method further comprises removing a sacrificial portion of a substrate comprising the device layer and the sacrificial portion to form the device layer.

(120) In one or more twenty-sixth embodiments, further to the twenty-fourth or twenty-fifth embodiments, the bonding layer is adjacent the device layer the method further comprises bonding a second support substrate to the interconnect layer, removing a sacrificial portion of a bulk substrate comprising the device layer and the sacrificial portion to form the device layer, and

separating the second support substrate from the interconnect layer.

(121) In one or more twenty-seventh embodiments, further to any of the twenty-fourth through twenty-sixth embodiments, forming the trench further comprises forming an opening within a perimeter defined by the trench, wherein the opening extends through the bonding layer, the interconnect layer, and the device layer, and exposes the support substrate, the method further comprising filling the trench with a via structure simultaneously with said filling the trench with the guard ring structure.

(122) In one or more twenty-eighth embodiments, further to any of the twenty-fourth through twenty-seventh embodiments, the method further comprises forming a second guard ring structure within the interconnect layer, wherein the second guard ring structure surrounds the metal interconnects and extends from a first surface of the interconnect layer and terminates at a first surface of the device layer, wherein the guard ring structure extends through the first surface of the interconnect layer, the first surface of the device layer, and a second surface of the device layer opposite the first surface of the device layer.

(123) In one or more twenty-ninth embodiments, further to any of the twenty-fourth through twenty-eighth embodiments, wherein the bonding layer is on a first surface of the support substrate and the guard ring structure extends not less than 200 nm beyond the first surface into support substrate.

(124) In one or more thirtieth embodiments, further to any of the twenty-fourth through twenty-ninth embodiments, at least one of the plurality of via structures is to couple to a ground plane.

(125) In one or more thirty-first embodiments, further to any of the twenty-fourth through thirtieth embodiments, the guard ring structure comprises a barrier layer and a fill material within the barrier layer, the barrier layer comprising a metal and oxygen and the fill material comprising copper.

(126) In one or more thirty-second embodiments, further to any of the twenty-fourth through thirty-first embodiments, the guard ring structure further comprises a polymer liner, wherein the barrier layer is within the polymer liner.

(127) In one or more thirty-third embodiments, further to any of the twenty-fourth through thirty-second embodiments, the support substrate and the device layer comprise silicon and the bonding layer comprises silicon and oxygen.

(128) In one or more thirty-fourth embodiments, further to any of the twenty-fourth through thirty-third embodiments, the guard ring structure comprises a lateral width between sidewalls thereof of not less than 4 microns.

(129) In one or more thirty-fifth embodiments, further to any of the twenty-fourth through thirty-fourth embodiments, the guard ring structure defines a substantially rectangular or square perimeter around the semiconductor devices and the metal interconnects.

(130) In one or more thirty-sixth embodiments, further to any of the twenty-fourth through thirty-fifth embodiments, the guard ring structure is laterally surrounded by outer edges of the device layer, the interconnect layer, the bonding layer, and the support substrate.

(131) It will be recognized that the invention is not limited to the embodiments so described, but can be practiced with modification and alteration without departing from the scope of the appended claims. For example, the above embodiments may include specific combination of features.

However, the above embodiments are not limited in this regard and, in various implementations, the above embodiments may include the undertaking only a subset of such features, undertaking a different order of such features, undertaking a different combination of such features, and/or undertaking additional features than those features explicitly listed. The scope of the invention should, therefore, be determined with reference to the appended claims, along with the full scope of equivalents to which such claims are entitled.

Claims

1. An apparatus, comprising: a device layer adjacent to an interconnect layer, the device layer comprising a plurality of semiconductor devices and the interconnect layer comprising a plurality of metal interconnects; a bond layer on a support substrate and adjacent one of the interconnect layer or the device layer; and a structure comprising metal, wherein the structure laterally surrounds the semiconductor devices and the metal interconnects, the structure extends through the bond layer, the interconnect layer, and the device layer, and the structure is in contact with the support substrate.
2. The apparatus of claim 1, wherein the bond layer comprises silicon and oxygen.
3. The apparatus of claim 1, wherein the bond layer is on a first surface of the support substrate and the structure extends not less than 200 nm beyond the first surface into support substrate.
4. The apparatus of claim 1, wherein the bond layer is adjacent the interconnect layer, the apparatus further comprising: a plurality of via structures within a perimeter of the structure, wherein each of the via structures extends through the bond layer, the interconnect layer, and the device layer, and is in contact with the support substrate.
5. The apparatus of claim 1, wherein the structure comprises a barrier layer and a fill material within the barrier layer, the barrier layer comprising a metal and oxygen and the fill material comprising copper.
6. The apparatus of claim 5, wherein the structure further comprises a polymer liner, wherein the barrier layer is within the polymer liner.
7. The apparatus of claim 1, further comprising: a heat exchanger thermally coupled to the support substrate.
8. An apparatus, comprising: a device layer adjacent to an interconnect layer, the device layer comprising a plurality of semiconductor devices and the interconnect layer comprising a plurality of metal interconnects; a layer on a support substrate and adjacent one of the interconnect layer or the device layer, the layer comprising silicon and oxygen; and a guard ring structure, wherein the guard ring structure laterally surrounds the semiconductor devices and the metal interconnects, extends through the layer, the interconnect layer, and the device layer, and is in contact with the support substrate.
9. The apparatus of claim 8, further comprising: a second guard ring structure within the interconnect layer, wherein the second guard ring structure surrounds the metal interconnects and extends from a first surface of the interconnect layer and terminates at a first surface of the device layer, wherein the guard ring structure extends through the first surface of the interconnect layer, the first surface of the device layer, and a second surface of the device layer opposite the first surface of the device layer.
10. The apparatus of claim 8, wherein the layer is on a first surface of the support substrate and the guard ring structure extends not less than 200 nm beyond the first surface into support substrate.
11. The apparatus of claim 8, wherein the layer is adjacent the interconnect layer, the apparatus further comprising: a plurality of via structures within a perimeter of the guard ring structure, wherein each of the via structures extends through the layer, the interconnect layer, and the device layer, and is in contact with the support substrate.
12. The apparatus of claim 8, wherein the guard ring structure comprises a barrier layer and a fill material within the barrier layer, the barrier layer comprising a metal and oxygen and the fill material comprising copper.
13. The apparatus of claim 8, further comprising: a heat exchanger thermally coupled to the support substrate.
14. An apparatus, comprising: an interconnect layer on a device layer, the device layer comprising a plurality of semiconductor devices and the interconnect layer comprising a plurality of metal interconnects; a bond layer between a support substrate and one of the interconnect layer or the device layer; and a guard ring structure, wherein the guard ring structure laterally surrounds the

semiconductor devices and the metal interconnects, extends through the bond layer, the interconnect layer, and the device layer, and is in contact with the support substrate.

15. The apparatus of claim 14, wherein the bond layer comprises silicon and oxygen.

16. The apparatus of claim 14, further comprising: a second guard ring structure within the interconnect layer, wherein the second guard ring structure surrounds the metal interconnects and extends from a first surface of the interconnect layer and terminates at a first surface of the device layer, wherein the guard ring structure extends through the first surface of the interconnect layer, the first surface of the device layer, and a second surface of the device layer opposite the first surface of the device layer.

17. The apparatus of claim 14, wherein the bond layer is on a first surface of the support substrate and the guard ring structure extends not less than 200 nm beyond the first surface into support substrate.

18. The apparatus of claim 14, wherein the bond layer is adjacent the interconnect layer, the apparatus further comprising: a plurality of via structures within a perimeter of the guard ring structure, wherein each of the via structures extends through the bond layer, the interconnect layer, and the device layer, and is in contact with the support substrate.

19. The apparatus of claim 14, wherein the guard ring structure comprises a barrier layer and a fill material within the barrier layer, the barrier layer comprising a metal and oxygen and the fill material comprising copper.

20. The apparatus of claim 14, further comprising: a heat exchanger thermally coupled to the support substrate.
